

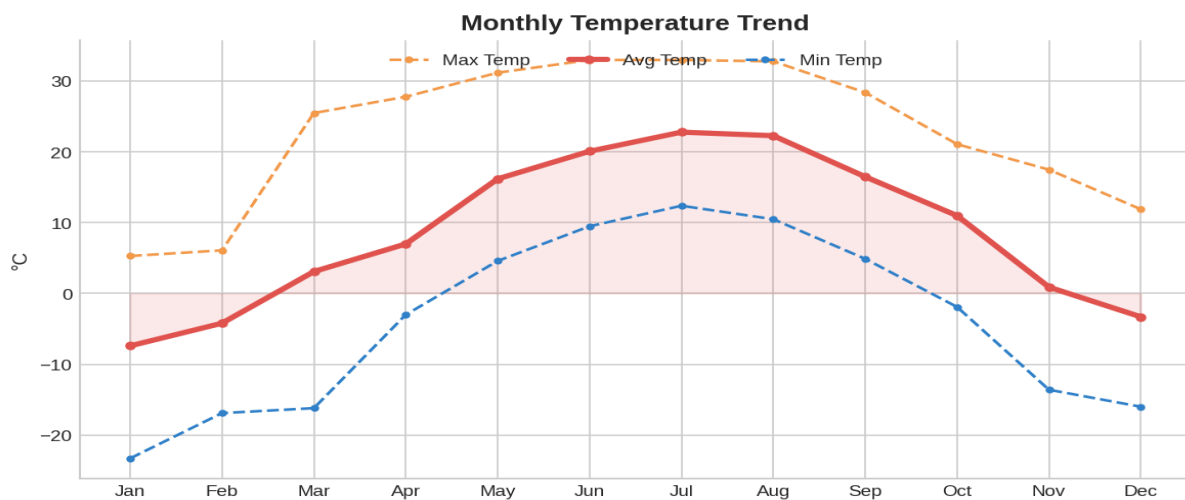
Toronto Weather Analytics — 2012



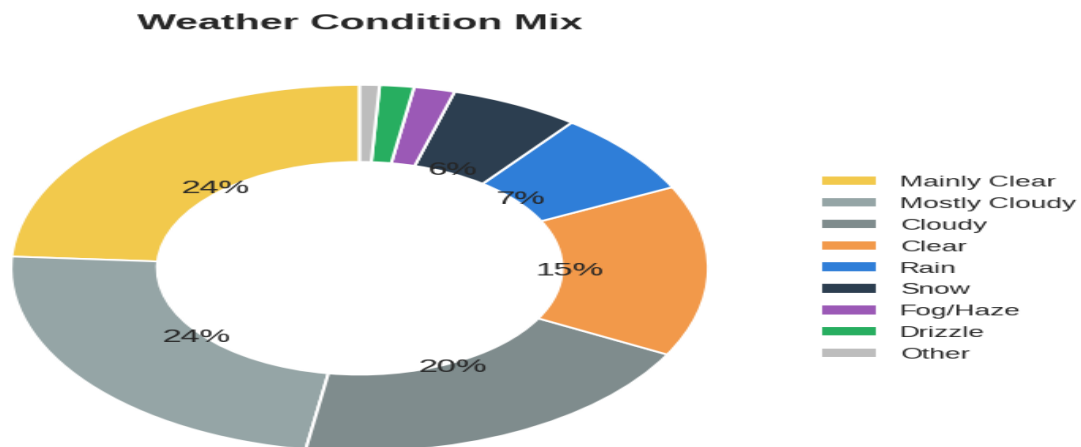
Hourly climate dataset · 8,784 observations · Jan 1 - Dec 31, 2012 (leap year)

Avg Temp	Max Temp	Min Temp	Avg Humidity	Avg Wind	Precip Hours
8.8°C	33.0°C	-23.3°C	67.4%	14.9 km/h	15.8%

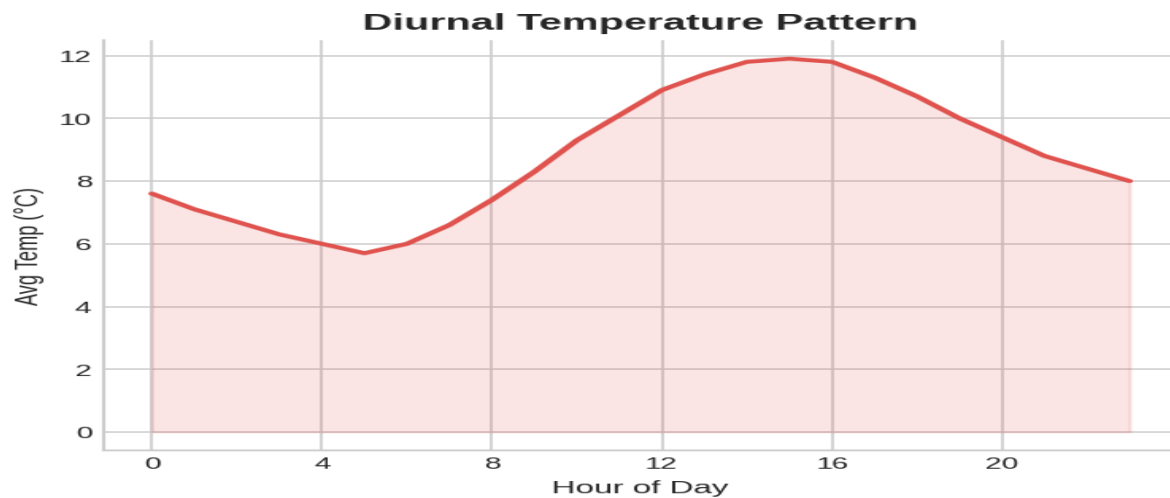
Monthly Temperature Trend



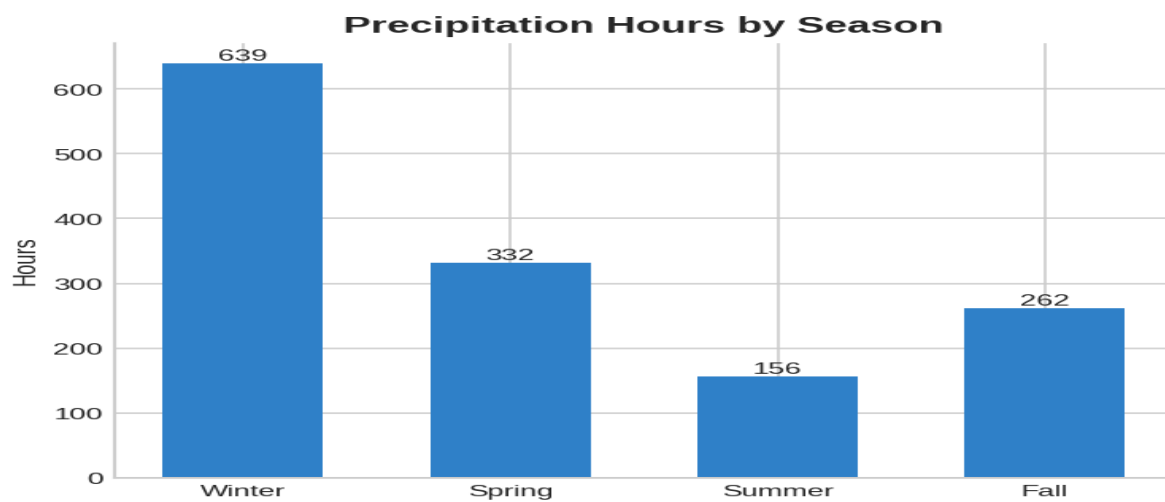
Weather Condition Mix



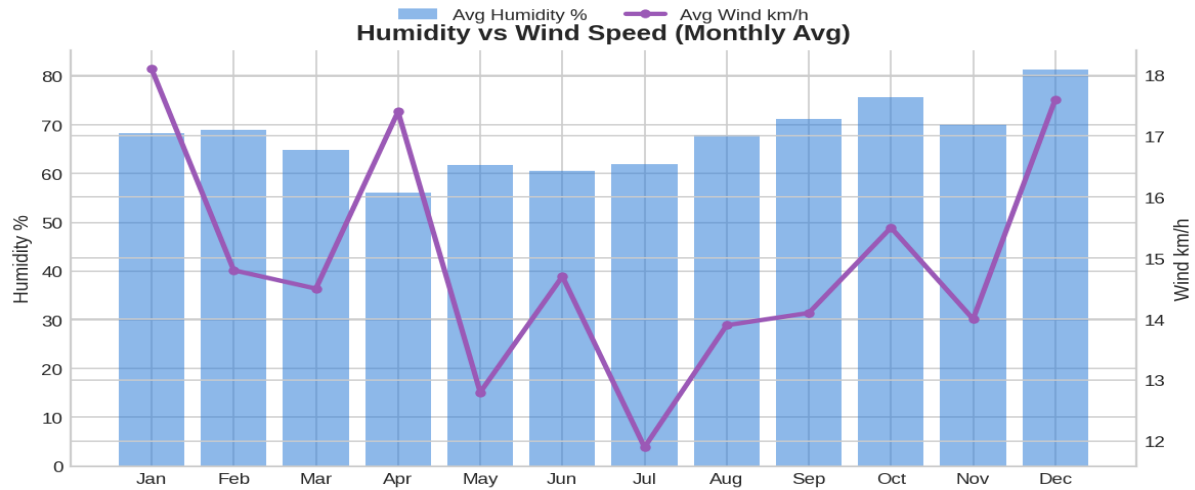
Diurnal Temperature Pattern



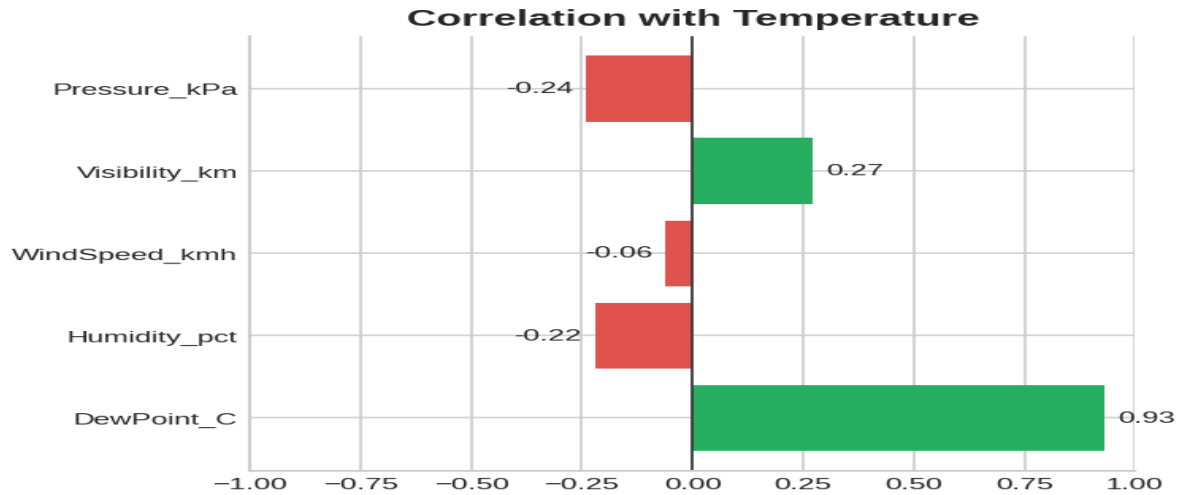
Precipitation Hours by Season



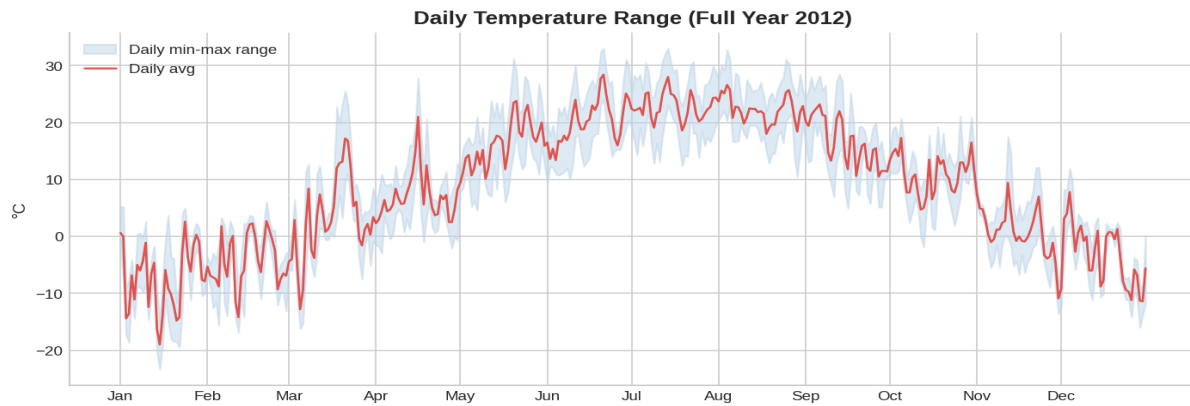
Humidity vs Wind Speed



Correlation with Temperature



Daily Temperature Range (Full Year 2012)



Monthly Summary

Month	Avg Temp	Max	Min	Humidity %	Wind km/h	Pressure kPa	Precip Hrs
Jan	-7.4	5.3	-23.3	68.4	18.1	101.0	221
Feb	-4.2	6.1	-16.9	69.0	14.8	101.1	150
Mar	3.1	25.5	-16.2	64.9	14.5	101.3	129
Apr	7.0	27.8	-3.0	56.2	17.4	100.7	100
May	16.2	31.2	4.6	61.8	12.8	101.1	103
Jun	20.1	33.0	9.5	60.6	14.7	100.8	83
Jul	22.8	33.0	12.4	62.0	11.9	100.8	30
Aug	22.3	32.8	10.5	67.9	13.9	100.9	43
Sep	16.5	28.4	4.9	71.2	14.1	101.1	74
Oct	11.0	21.1	-1.9	75.7	15.5	100.9	100
Nov	0.9	17.5	-13.6	70.0	14.0	101.7	88
Dec	-3.3	11.9	-16.0	81.3	17.6	101.2	268

Analysis Summary

- **Strong seasonality:** average temperature swings from about -7.4°C in January to roughly $21\text{-}22^{\circ}\text{C}$ in summer, an annual range of 56.3°C between the single hottest (33.0°C , June) and coldest (-23.3°C , January) hours recorded.
- **Predominantly dry/clear conditions:** roughly 39.1% of all hours were clear or mainly clear, while only 15.8% had active precipitation (rain, snow, drizzle, or freezing variants) and 5.2% had fog/haze.
- **Winter is the wettest/foggiest season** by hour-count (highest precipitation hours), while summer is driest despite being warmest — typical of a continental pattern where winter precipitation is more persistent (snow) even if less intense.
- **Humidity and wind are inversely related to visibility:** humidity correlates at $r \approx -0.63$ with visibility, and pressure drops are associated with higher wind speeds ($r \approx -0.36$) — both physically expected relationships that validate data integrity.
- **Dew point tracks temperature almost perfectly** ($r \approx 0.93$), as expected physically, while pressure and temperature show a mild negative relationship, reflecting more low-pressure (stormy) systems arriving in cooler months.
- **Diurnal cycle:** temperatures trough in the early morning ($\sim 5\text{-}6$ AM) and peak in mid-afternoon ($\sim 3\text{-}4$ PM), a normal solar heating pattern.

Data quality: the dataset was already clean — zero nulls, zero duplicate timestamps, complete hourly coverage across the full leap year, and no out-of-range values (humidity 0-100%, non-negative wind/visibility). Cleaning consisted of standardizing column names, parsing the datetime field, and deriving categorical/time features (season, hour, primary weather condition, precipitation/fog flags) to support this analysis.

Source: Environment Canada hourly climate station data (2012). Report generated with Python (pandas, matplotlib, reportlab). · dabelinfotech